

Configuration Manual

MSc Research Project
MSc in Cloud Computing

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Project Submission Sheet
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Configuration Manual

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This document provides a detailed configuration guide for setting up, deploying, and running the Mental Health Analysis project. The steps cover all aspects, from library installation to deploying the project on AWS using services such as Cloud9, S3, DynamoDB, Elastic Beanstalk, and CodePipeline.

1. Installing Required Libraries

To ensure all dependencies are installed, run the following commands in your terminal. These commands install the required Python libraries for the project (Pedregosa et al.; 2011):

- `pip install flask`
- `pandas`
- `numpy`
- `scikit-learn`
- `boto3`
- `awscli`
- `matplotlib`
- `seaborn`
- `nltk`
- `vaderSentiment`
- `markovify`
- `SentimentIntensityAnalyzer`

2. Setting up AWS Cloud9 Environment

1. Log in to the AWS Management Console.(Amazon Web Services; 2024d)
2. Navigate to **Cloud9** under Developer Tools.
3. Click on **Create Environment**.

4. Enter a name (e.g., *MHA-Development*) and provide a description.
5. Choose the instance type (e.g., *t2.micro*) and configure other options.
6. Click **Create** to launch the environment.
7. In the Cloud9 terminal, ensure Python and pip are updated by running:

```
sudo yum update -y
sudo pip install --upgrade pip
```

3. Setting up an S3 Bucket

1. Navigate to the AWS Management Console and go to **S3** (Amazon Web Services; 2024c).
2. Click **Create Bucket**.
3. Provide a unique bucket name (e.g., *mha-dataset*).
4. Select your desired region and configure the public access settings (ensure sensitive data remains private).
5. Click **Create Bucket**.
6. Upload the dataset files:
 - (a) Click on the bucket name.
 - (b) Click **Upload** and drag the dataset files into the interface.
7. Note the bucket name and region for later use.

4. Creating a DynamoDB Table

1. Navigate to **DynamoDB** in the AWS Management Console (Amazon Web Services; 2024b).
2. Click **Create Table**.
3. Enter the table name (e.g., *MHAResults*).
4. Set the partition key to **user_id** (String).
5. Leave the other settings as default and click **Create Table**.
6. Once created, note the table name for use in your Flask application.

5. Setting up Elastic Beanstalk

1. Go to **Elastic Beanstalk** in the AWS Management Console (Amazon Web Services; 2024f).
2. Click **Create Application**.
3. Provide an application name (e.g., *MHApp*).

4. Choose the platform as **Python**.
5. Upload your Flask project ZIP file (`MHA_Flask.zip`) or deploy a sample application.
6. Click **Create Application**.
7. To scale instances:
 - (a) Go to the application's **Configuration**.
 - (b) Under **Capacity**, set the desired number of instances for auto-scaling.

6. Setting up CodePipeline

1. Navigate to **CodePipeline** in the AWS Management Console (Amazon Web Services; 2024e).
2. Click **Create Pipeline**.
3. Provide a pipeline name (e.g., *MHADeploymentPipeline*).
4. Under **Source**, select GitHub, authenticate, and choose your repository.
5. Under **Build**, select AWS CodeBuild or skip if not required.
6. Under **Deploy**, choose **Elastic Beanstalk** and select your application.
7. Click **Create Pipeline**.

7. Using Git for Code Changes

To manage and push changes to the repository, use the following commands:

```
git init
git add .
git commit -m "Initial -commit"
git remote add origin <your-repository-url>
git push -u origin main
```

To push updates and trigger the pipeline:

```
git add .
git commit -m "Updated -model"
git push
```

8. Configuring CloudWatch

1. Navigate to **CloudWatch** in the AWS Management Console (Amazon Web Services; 2024a).
2. Click **Alarms** *;* **Create Alarm**.
3. Choose a metric related to Elastic Beanstalk, such as **CPUUtilization**.

4. Set thresholds and configure notification actions (e.g., email alerts).
5. Save and enable the alarm.

9. Installing Jupyter Notebook Using Anaconda

9.1 Jupyter Notebook

Jupyter Notebook is an open-source web application that allows you to create and share documents containing live code, equations, visualizations, and narrative text. Follow these steps to install and set up Jupyter Notebook using Anaconda:

1. **Download Anaconda:** Visit the official Anaconda website at <https://www.anaconda.com/> and download the Anaconda distribution suitable for your operating system (Windows, macOS, or Linux).
2. **Install Anaconda:** (Anaconda; 2023)
 - For Windows: Run the installer and follow the on-screen instructions. Ensure you check the option to add Anaconda to your PATH environment.
 - For macOS and Linux: Open the downloaded installer file and follow the installation prompts in the terminal.
3. **Verify Installation:** Open a terminal or command prompt and type:

```
conda --version
```

This command should return the installed version of Conda.

4. **Launch Jupyter Notebook:** (Jupyter; 2023) After installation, launch Jupyter Notebook by running the following command in the terminal or Anaconda Prompt:

```
jupyter notebook
```

This will open the Jupyter Notebook interface in your default web browser.

5. **Create a New Notebook:** Click **New** in the top-right corner and select **Python 3** to create a new Python notebook.

9.2 Setting up Google Colab

Google Colab is a cloud-based notebook environment that allows you to write and execute Python code in an interactive, collaborative way. To set up Google Colab, follow these steps (Colaboratory; 2023):

1. **Access Google Colab:** Open your web browser and navigate to <https://colab.research.google.com/>.
2. **Sign In with Google Account:** Log in using your Google account. If you do not have one, create a Google account at <https://accounts.google.com/>.

3. **Create a New Notebook:** Once logged in, click on **File > New Notebook** to create a new Colab notebook. Alternatively, click the **+ New Notebook** button on the homepage.
4. **Upload Notebook Files:** If you want to run an existing notebook file, click **File > Upload Notebook** and upload the required `.ipynb` file.
5. **Install Libraries:** Install any additional Python libraries using the `!pip install` command in a code cell. For example:

```
!pip install pandas
```

6. **Save and Share:** Google Colab automatically saves your work in your Google Drive. To share the notebook, click **Share** in the top-right corner and configure sharing permissions.

10. Running Notebooks

1. Open the `MHA.ipynb` notebook in Jupyter Notebook or Google Colab.
2. Execute all cells to preprocess the dataset.
3. Open `Model_Training.ipynb` and execute all cells to train and evaluate the machine learning models. .

11. Running the Flask Application

1. Navigate to the directory containing the Flask application (`MHA_Flask`) (Projects; 2024) (Python Software Foundation; 2024).
2. Install the required dependencies from `requirements.txt`:

```
pip install -r requirements.txt
```

3. Run the Flask application locally:

```
python app.py
```

4. Test the application by visiting `http://127.0.0.1:5000` in your browser.

12. Configuring Load Balancing

1. Go to your Elastic Beanstalk application.
2. Under **Configuration**, navigate to **Load Balancer** (Services; 2023).
3. Enable load balancing and configure the listener settings.
4. Save and apply the changes.

Conclusion

By following this configuration guide, you can successfully set up and run the MHA project. This document ensures reproducibility and scalability, leveraging the power of AWS services and Python-based development tools.

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